Alex Gan

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Education

University of Pennsylvania **Philadelphia, PA**
B.S.E in Computer Science, University Scholar Expected May 2030
Current Coursework: Multivariable Calculus, Programming Languages and Techniques, Mathematical Foundations of Computer Science

McLean High School **McLean, VA**
High School Diploma June 2026
GPA: 4.7 (weighted) | SAT: 1570
Relevant Coursework: Linear Algebra, Differential Equations, Discrete Math, AP Physics C (Mechanics & E&M), AP Computer Science A, AP Calculus BC, AP Statistics

Experience

2 Times Aspiring Scientists Summer Internship Program (ASSIP) (2024, 2025)

- Researched under Dr. Mariia Petryk & Dr. Li Jiasun (Finance, Costello College of Business, GMU), analyzing open-source blockchain projects using SQL and Python.
- Analyzing the Impact of Different Funding and Governance Models on an OSS Project's Success
- Empirical Validation of Account Abstraction: A Quantitative Analysis of Transaction Performance and Adoption Patterns in the Ethereum Blockchain

Estimating Hidden Housing Insecurity Beyond Point-in-Time Counts (2025-2026)

- Built Bayesian evidence-synthesis and Monte Carlo simulation models to estimate housing insecurity not captured by official Point-in-Time counts.
- Presented findings at the 2026 New England Statistics Symposium.

Correlation of Air Pollution with Population Density and Traffic Conditions Using Google Earth Engine and Predictive Models (2024–2025)

- Applied Google Earth Engine and machine learning (XGBoost) to study relationships between NO_2 pollution, population density, and traffic patterns, enabling scalable and reproducible urban air-quality monitoring.

Co-Captain, First Tech Challenge, Double Quarter Pounders (2023–2026)

- Created 6 functioning robots. Programmed the robots autonomously using Java.
- Created programming lessons for new team members and younger students attending our summer camp.

GStack x GBrain Hackathon, Y Combinator (2026)

- Created Oncura, a service for clinical trial sites to increase patient recruitment efficiency.
- Utilized Claude Code, Codex, Rust, C++, and GitHub to build and organize Oncura.

Skills

Programming: Python, Java, JavaScript, HTML, CSS, SQL, OCaml
Technical Skills: Data Structures & Algorithms, NumPy, Matplotlib, OpenCV, MongoDB, Machine Learning, Data Analysis & Visualization
Languages: English (Native), Chinese (Fluent)